

AI-Powered Document Classification System

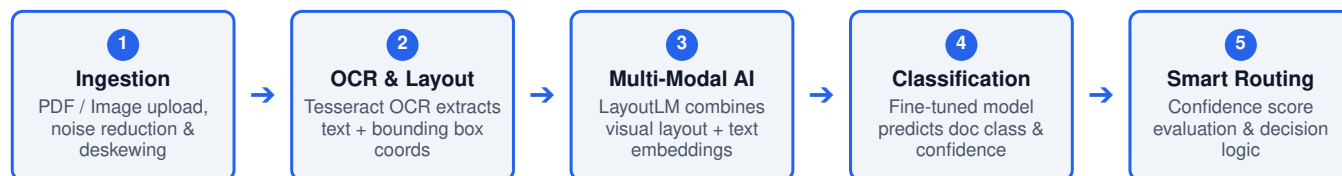
Prepared by: Vedant Ajay Ware

Department: Applied AI & NLP (CASAS)
Academic Level: Third Year / Final Year Degree
Domain: Intelligent Automation & OCR

EXECUTIVE SUMMARY & CORE OBJECTIVE

Traditional document processing relies on rigid rules and manual data entry, resulting in high error rates and operational bottlenecks. This project presents an end-to-end **Layout-Aware AI Document Processing System** that automatically ingests unstructured digital paperwork (invoices, contracts, IDs, forms), extracts text alongside visual-spatial context, classifies documents using multi-modal deep learning, and intelligently routes them for enterprise workflow automation.

SYSTEM ARCHITECTURE & PROCESSING WORKFLOW



High Confidence ($\geq 85\%$) Straight-Through Processing (STP)

Automated metadata extraction and direct integration into downstream enterprise backend database / ERP systems without human delay.

Low Confidence ($< 85\%$) Human-In-The-Loop (HITL) Review

Flagged for manual review on Streamlit / React Web Dashboard with visual bounding-box feedback for quick operator verification.

KEY TECHNICAL CAPABILITIES

- Multi-Modal Feature Fusion:** Analyzes visual document layout, spatial typography, and semantic text simultaneously using Hugging Face LayoutLM.
- Layout-Aware OCR:** Preserves positional bounding boxes (x_0, y_0, x_1, y_1) ensuring precise spatial understanding across variable templates.
- Hybrid Decision Pipeline:** Combines fine-tuned deep learning models for standardized classes with LLM fallback for complex unstructured documents.
- Interactive Admin Dashboard:** Provides real-time visual inspection, document drag-and-drop, and analytical performance metrics.

RECOMMENDED TECH STACK

Layer	Technologies
Languages & Processing	Python 3.10+, OpenCV, Tesseract OCR
AI / ML Core	PyTorch, Hugging Face, LayoutLM / BERT
Backend & API	FastAPI (RESTful services), Pydantic
Frontend UI	Streamlit / React.js Interactive Dashboard
Datasets Used	RVL-CDIP (400k benchmark images), Kaggle

92%+

TARGET F1-SCORE ACCURACY

< 1.0 Sec

CLASSIFICATION LATENCY / DOC

16 Classes

BENCHMARK DOCUMENT TYPES